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ODD END SEMESTER EXAMINATION DECEMBER- 2024

Name of The Program: **B.Tech (Petroleum Engineering)**
Name of The Course: **Advanced Well Stimulation Technique**

Semester: **VII**
Course Code: **TPE-723**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are Compulsory
- (ii) Answer any two sub questions among a, b & c in each Main Question.
- (iii) Total marks in each main question are twenty
- (iv) Each question Carries 10 marks.

Q1.

(20 Marks) (CO1,2)

- a) Explain the following causes of skin
 - i) Solids plugging
 - ii) Clay-particle swelling or dispersion
 - iii) Saturation changes
 - iv) Wettability reversal
 - v) Emulsion blockage
 - vi) Aqueous-filtrate blockage
- b) What are different effects of reservoir properties & Effects of perforations on fracture execution.
- c) Explain Hydrofracturing and mechanism for fracture generation.

Q2.

(20 Marks) (CO2,3)

- a) Write a short note on
 - i) Mineral Acid
 - ii) Retarded Acid
 - iii) Organic Acid
- b) Name and explain the various additives that are used to enhance acidification performance.
- c) What are the various environmental and operational factors to be taken into account during acidization.

Q3.

(20 Marks) (CO3, 4)

- a) What does "Formation Type" and "Formation Permeability" actually mean? Give a brief explanation.
- b) Explain skin, including its positive, negative, and zero aspects.
- c) Write different operation procedure of Hydrofracturing.

Q4.

(20 Marks) (CO4,5)

- a) Explain what is meant by "acidization," the various techniques that can be used, and the differences between any two of them.

- b) Define proppant. What are different types of proppants used for hydraulic fracturing and also explain how proppant selection is done.
- c) Which acids are used for the following issues, explain with their primary reactions.
 - i) Clay Blockage
 - ii) Carbonate Blockage
 - iii) Corrosion

Q5

(20 Marks) (CO3, 4)

- a) Describe formation damage and prevention of formation damage.
- b) List the various formation damage mechanisms and provide a brief description of each.
- c) What are different considerations/parameters in fracture design.